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To: *ACM Computing Surveys (CSUR)* editorial board.

Dear editors,

Please find attached our manuscript entitled "Time-series dimension reduction: a comprehensive review and conceptual unification of algorithms" that we are submitting for publication in *CSUR* for consideration as a *long survey paper*.

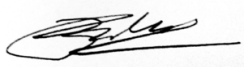
With the ubiquity of high-dimensional multivariate time-series data across diverse modern applications, dimension-reduction (DR) algorithms have become indispensable for extracting informative lower-dimensional time-series components which facilitate characterization, modeling, and prediction of these complex datasets. But the most commonly used DR methods—including principal component analysis (PCA) and autoencoder (AE) neural networks—are designed for general high-dimensional data and do not account for the temporal structure of time-series data. However, there exist scores of powerful but lesser-known *time-series dimension reduction* (TSDR) methods have been developed that do take temporal structure into account. Remarkably, a recent survey of DR techniques *for time-series data* (Ashraf et al., 2023, *IEEE Access*) encompassed only general DR techniques but **did not include *any* of TSDR methods that account for temporal ordering**. This underscores the level of underappreciation for the rich and interdisciplinary TSDR literature, even amongst experts in applying dimension reduction to time series.

A variety of powerful TSDR methods exist, developed across a range of disciplines, but they remain drastically underappreciated and have never previously been reviewed or unified.

In this paper we comprehensively synthesize the literature on time-series dimensional reduction methods for the first time, encompassing over sixty TSDR methods, providing the first unified resource for analysts to learn about this rich methodological literature. We also introduce a conceptual organization of these TSDR methods across six categories, outlining the core motivation and algorithmic steps involved in each category of methods. This allows us to then map and explain relationships between the various TSDR methods, providing a unified organization of an interdisciplinary literature. This includes new equivalence results between disparate TSDR methods that we demonstrate here for the first time. Finally, we provide an accompanying python software package, *tsdr*, to facilitate the uptake, usage, and further development of TSDR methods, and demonstrate the power of these methods on simulated data relative to conventional DR methods.

Our accessible and unifying review of a drastically underappreciated methodological literature will be essential reading for researchers in theory and applications of time-series analysis and machine learning. By providing an associated software package, we anticipate that this work will become a landmark paper, providing a foundational reference for the growing community empirical scientists analyzing high-dimensional time-series data. We propose that our paper best suits the *long survey paper* type, since it includes a review and perspective on the field, and accompanying statistical software. Thank you for your consideration of our work.

Sincerely,



A/Prof. Ben D. Fulcher, on behalf of the authors